PROJECT AND TEAM INFORMATION

Project Title

VIRTUAL CITY SIMULATOR

Student / Team Information

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PROPOSAL DESCRIPTION (10 pts)

Motivation

Modern cities have many interconnected systems such as roads, buildings, vehicles, traffic, emergency services, water, electricity, and finance. Managing these systems efficiently can be complex, especially when traffic congestion, emergencies, or limited resources occur. Our project Virtual City Simulator aims to provide a simplified virtual model of a city where these components can be managed and monitored in one system.

The project is important because it demonstrates how Data Structures and Algorithms (DSA) can be applied to solve real-world city-management problems. The road network is represented as a weighted graph, where intersections are nodes and roads are edges. Dijkstra's algorithm is used to find the shortest routes for vehicles, ambulances, and food-delivery services. Traffic conditions can be managed using queues and priority queues based on the number of waiting vehicles and their waiting time.

Through this simulator, we aim to understand how DSA and OOP concepts can work together to model and manage a complex system. The project provides a terminal-based interface where users can manage and monitor different components of the virtual city through menu-driven operations.

State of the Art / Current solution

Today, city management is generally handled through Intelligent Transportation Systems (ITS) and smart-city technologies. These systems use sensors, cameras, real-time traffic information, intelligent traffic lights, and centralized control systems to monitor traffic and improve transportation efficiency. Modern systems can also support dynamic route planning and emergency-vehicle prioritization.

For emergency services, dedicated systems use GIS, automated dispatching, vehicle tracking, and shortest-path analysis to help emergency vehicles reach their destinations quickly.

Our current project provides a simplified simulation of these concepts. The city road network is represented as a weighted graph, with intersections as nodes and roads as edges. The system plans routes using Dijkstra's shortest-path algorithm and includes separate operations for ambulances and food-delivery vehicles.

The project also models traffic monitoring using vehicle waiting time and a priority queue, while providing sections for buildings, resources, and city finance.

Thus, our solution is a small-scale educational simulation of real-world smart-city management using DSA and OOP concepts.

Project Goals and Milestones

The main goal of the Virtual City Simulator is to develop a city-management system that demonstrates the practical application of Data Structures, Algorithms, and Object-Oriented Programming (OOP). The simulator will model important city components such as roads, intersections, buildings, population, vehicles, traffic, emergency services, resources, and finance. The project will provide a menu-driven terminal interface to monitor and manage these components.

Step-by-Step Milestones

- ***Requirement Analysis*** *– Identify the city components and finalize the functional requirements of the simulator.*
- ***Menu Design*** *- Design a terminal-based menu system and organize options for roads, buildings, vehicles, traffic, resources, and finance.*
- ***Road Network Implementation*** *– Create intersections and roads using a weighted graph, where intersections are nodes and roads are edges.*
- ***Building & Population Module*** *– Implement houses, hospitals, restaurants, and population management.*
- ***Vehicle Module*** *– Add cars, ambulances, and food-delivery bikes with their locations and basic information.*
- ***Shortest-Path Module*** *– Implement Dijkstra's algorithm to find the shortest route between two intersections.*
- ***Emergency & Delivery Module*** *– Use shortest-path routing for ambulance dispatch and food delivery.*
- ***Traffic Management*** *– Implement vehicle waiting queues, waiting-time monitoring, and priority-based traffic management.*
- ***Resources & Finance*** *– Add water, electricity, city funds, expenses, and resource-purchase management.*
- ***Integration, Testing & Final Demonstration*** *– Combine all modules, test the complete system, fix errors, and prepare the final PBL demonstration and documentation.*

Project Approach

We plan to develop the Virtual City Simulator as a terminal-based C++ city-management system. The project follows a modular approach, where each city component is developed and tested separately before integrating everything into the final system.

First, we will design a simple menu-driven terminal interface that allows users to access and manage city information such as intersections, buildings, population, vehicles, traffic, resources, and finance.

Next, the city road network will be designed using a weighted graph, where intersections represent nodes and roads represent edges. An adjacency list will be used to efficiently store road connections. Buildings and population will then be connected to the city network.

The vehicle module will support cars, ambulances, and food-delivery bikes. Dijkstra's algorithm will be implemented to calculate the shortest path between intersections. This will be used for ambulance dispatch and food delivery.

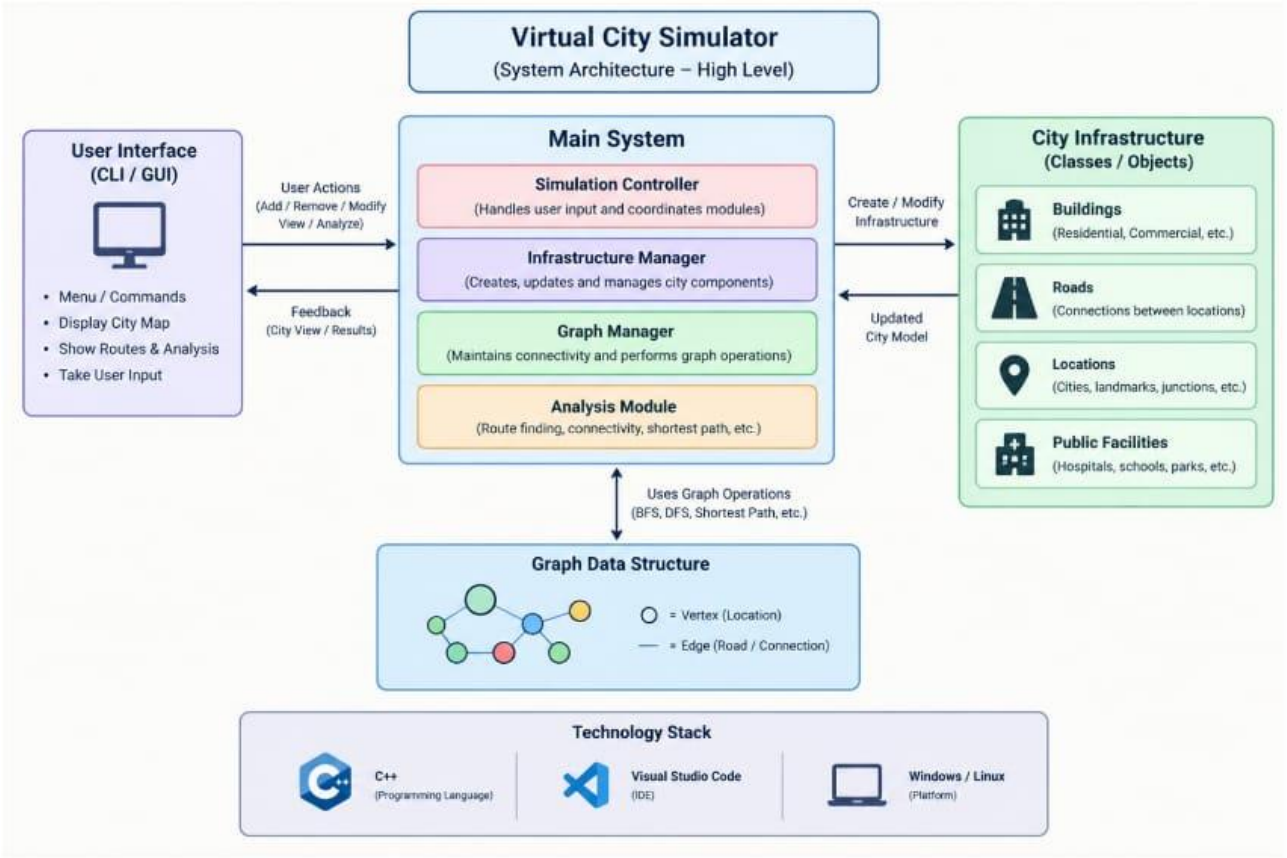
For traffic management, queues and priority queues will be used to manage waiting vehicles and assign traffic priority based on waiting conditions.

Finally, water/electricity resources and city finance will be integrated. Each module will be tested individually, followed by system integration, debugging, and final testing to produce a complete working simulator.

System Architecture (High Level Diagram)

- ***Terminal Interface*** *- Allows the user to interact with the simulator through a menu-driven console and manage roads, buildings, population, vehicles, traffic, resources and finance.*

- ***Building & Population Module-*** *Manages houses, hospitals and restaurants, along with the population living in houses.*
- ***Vehicle Management Module-*** *Maintains different types of vehicles, including cars, ambulances and food-delivery bikes, along with their current locations.*
- ***Route Optimization Module-****Uses Dijkstra's algorithm to find the shortest path between two intersections for efficient vehicle movement.*
- ***Emergency & Delivery Module-****Uses the shortest route to dispatch ambulances to accident locations and deliver Food from restaurants to customers' houses.*
- ***Traffic Management Module-****Monitors vehicles waiting at intersections and manages traffic priority using queues and priority queues based on waiting conditions.*
- ***Resources & Finance Module-*** *Monitors water and electricity availability and manages city funds, expenses and resource purchases.*



Project Outcome / Deliverables

The expected outcome of the project is a working Virtual City Simulator that demonstrates city management, route optimization, vehicle management and traffic control using DSA and OOP concepts.

The system will allow users to view and manage different components of a virtual city, including roads, intersections, buildings, population and vehicles. It will represent the road network using a weighted graph and use Dijkstra's algorithm to find the shortest routes. Vehicles such as cars, ambulances and food-delivery bikes will be managed within the system.

Expected Deliverables

- *Working C++ implementation of the core system.*
- *Interactive terminal-based menu system.*
- *Road network and intersection management.*
- *Weighted graph and adjacency-list implementation.*
- *Building and population management module.*
- *Vehicle management module.*
- *Dijkstra's shortest-path implementation.*
- *Ambulance emergency-response system.*
- *Food-delivery route management.*
- *Traffic monitoring and management using queues and priority queues.*
- *Water and electricity resource management.*
- *City finance and expense management.*
- *Test cases and testing results.*
- *Project documentation.*
- *Final project demonstration.*

Assumptions

- *The city is represented as a finite weighted graph consisting of intersections and roads.*
- *Vehicles are assumed to move between connected intersections.*
- *Dijkstra's algorithm is used only when road weights are non-negative.*
- *The number of vehicles, buildings, and population is manageable for the simulation.*
- *Traffic priority is determined using the number of waiting vehicles and their waiting time.*
- *Water and electricity requirements are assumed to have predefined values.*
- *The simulator represents a simplified model of a real city for educational purposes.*

References

- *C++ Reference – cppreference.com — Used for C++ concepts and standard library data structures such as queues and priority queues.*
- *Dijkstra's Algorithm – Wikipedia — Used for understanding the shortest-path algorithm used in the road network.*
- *Graph Data Structure – TutorialsLogic — Used for understanding graph representation, vertices, edges, and adjacency lists.*
- ***Project Source Code** — Used as the basis for designing and implementing the Virtual City Simulator modules, including road network, vehicles, route optimization, emergency response, traffic monitoring, resources, and finance.*